Cloud-Based Active Directory Setup & User Management using Microsoft Entra ID

Project Overview

This project demonstrates the implementation of a cloud-based Identity and Access Management (IAM) environment using Microsoft Entra ID (formerly Azure Active Directory). The objective was to create a secure identity infrastructure capable of managing users, groups, roles, authentication methods, and access lifecycle processes while following cybersecurity best practices.

The project was built using a Microsoft Azure Free Tier subscription and focuses on core IAM concepts used in modern enterprises.

Objectives

- Create and manage cloud identities.
- Implement group-based access management.
- Apply Role-Based Access Control (RBAC).
- Configure authentication methods.
- Monitor identity-related activities through audit logs.
- Simulate employee lifecycle management (Joiner, Mover, Leaver).
- Follow the Principle of Least Privilege.

Technologies Used

- Microsoft Azure
- Microsoft Entra ID (Azure Active Directory)
- Azure Portal
- Audit Logs
- Role-Based Access Control (RBAC)

Architecture

Users | v Microsoft Entra ID Tenant | +-----------------------+ | | v v Security Groups Roles | | v v Access Control Permissions | v Authentication Methods | v Audit Logs & Monitoring

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Implementation Steps

1. Tenant Setup

- Created Microsoft Azure Free Tier account.
- Accessed Microsoft Entra ID tenant.

- Explored tenant properties and identity management features.

2. User Provisioning

Created multiple cloud identities:

- employee1
- developer.user
- security.analyst
- intern.user

Activities performed:

- User creation
- User management
- Account status management

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3. Security Group Creation

Created department-based security groups:

- IT-Team
- HR-Team
- Developers-Team
- Security-Team
- Finance-Team

Purpose:

- Simplified permission management
- Improved scalability
- Enabled group-based access control

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4. Role-Based Access Control (RBAC)

Assigned built-in Microsoft Entra roles.

Example:

Role Assigned:

- User Administrator

Benefits:

- Delegated administration
- Reduced dependency on Global Administrators
- Improved security through least privilege

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5. Authentication Management

Configured authentication methods for users.

Security benefits:

- Improved account protection
- Support for Multi-Factor Authentication (MFA)
- Enhanced identity verification

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6. Audit Logging

Monitored activities using Microsoft Entra Audit Logs.

Observed events:

- User creation
- Group creation
- User added to group
- User removed from group
- Role assignment changes
- Account disable operations

Purpose:

- Security monitoring
- Compliance auditing
- Incident investigation
- Change tracking

Joiner-Mover-Leaver (JML) Lifecycle Management

Joiner Process

Scenario: New employee joins the organization.

Actions:

- Created security.analyst user
- Added user to Security-Team

Result:

- User successfully provisioned.

Mover Process

Scenario: Employee changes department.

Actions:

- Removed developer.user from Developers-Team
- Added developer.user to Security-Team

Result:

- Access adjusted according to new responsibilities.

Leaver Process

Scenario: Employee leaves organization.

Actions:

- Disabled intern.user account
- Verified audit log generation

Result:

- Access revoked successfully.

Cybersecurity Concepts Demonstrated

Identity and Access Management (IAM)

Managed identities, authentication, and authorization.

Principle of Least Privilege

Users received only the minimum access required.

Role-Based Access Control (RBAC)

Permissions managed using predefined roles.

Identity Governance

Implemented Joiner-Mover-Leaver workflows.

Security Monitoring

Tracked administrative activities using audit logs.

Account Lifecycle Management

Managed user creation, modification, and deactivation.

Challenges Encountered

Sign-in Log Access Restrictions

Issue: Unable to access Sign-in Logs due to permission/licensing limitations.

Resolution: Used Audit Logs to perform activity monitoring and auditing.

Premium Feature Limitations

The following features required Microsoft Entra Premium licensing:

- Conditional Access
- Self-Service Password Reset (SSPR)
- Advanced Identity Protection

Project continued using Free Tier capabilities.

Key Learnings

- Microsoft Entra ID fundamentals
- User and group management
- RBAC implementation
- Authentication management
- Audit logging and monitoring
- Identity lifecycle management
- Cloud IAM security principles

Future Enhancements

- Conditional Access Policies
- Self-Service Password Reset (SSPR)
- Privileged Identity Management (PIM)
- PowerShell Automation
- Log Analytics Integration
- Microsoft Sentinel Integration
- Identity Governance Workflows

Conclusion

This project successfully implemented a cloud-based identity and access management environment using Microsoft Entra ID. The solution demonstrated core cybersecurity concepts including user provisioning, access control, authentication management, audit logging, and lifecycle management. The project provides hands-on experience with enterprise IAM practices commonly used in cloud security and cybersecurity operations.